

IVIG for Neonatal Alloimmune Haemolytic Disease

Evidence at a glance — systematised narrative review, September 2026
Imperial College Healthcare NHS Trust — Queen Charlotte's & Chelsea and St Mary's Neonatal Units

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Placebo-controlled, double-blind Rh trials under modern intensive phototherapy found any benefit of IVIG on exchange-transfusion rate.

Smits-Wintjens 2011 (PMID 21422084); Santos 2012 (PMID 22882285)

69% → 15%

Fall in control-arm exchange-transfusion rate from the earliest (1992) to the most recent (2012) Rh trial — explaining most of the apparent "IVIG effect" in older studies.

Domain A / cross-cutting.md §1

31.66 → null

The largest adjusted odds ratio for IVIG and necrotising enterocolitis (Figueras-Aloy 2009) did not replicate in the one study designed to control for confounding by indication (Li et al. 2022, n=1259).

PMID 19948572; PMID 35613867

No increase

In late anaemia or top-up transfusion requirement attributable to IVIG — both placebo-controlled RCTs found no significant difference vs. control.

Domain E (e)

0 trials

Comparative studies of IVIG in Kell, anti-c, anti-E or DAT-negative ABO disease. No direct evidence exists for these groups on any outcome.

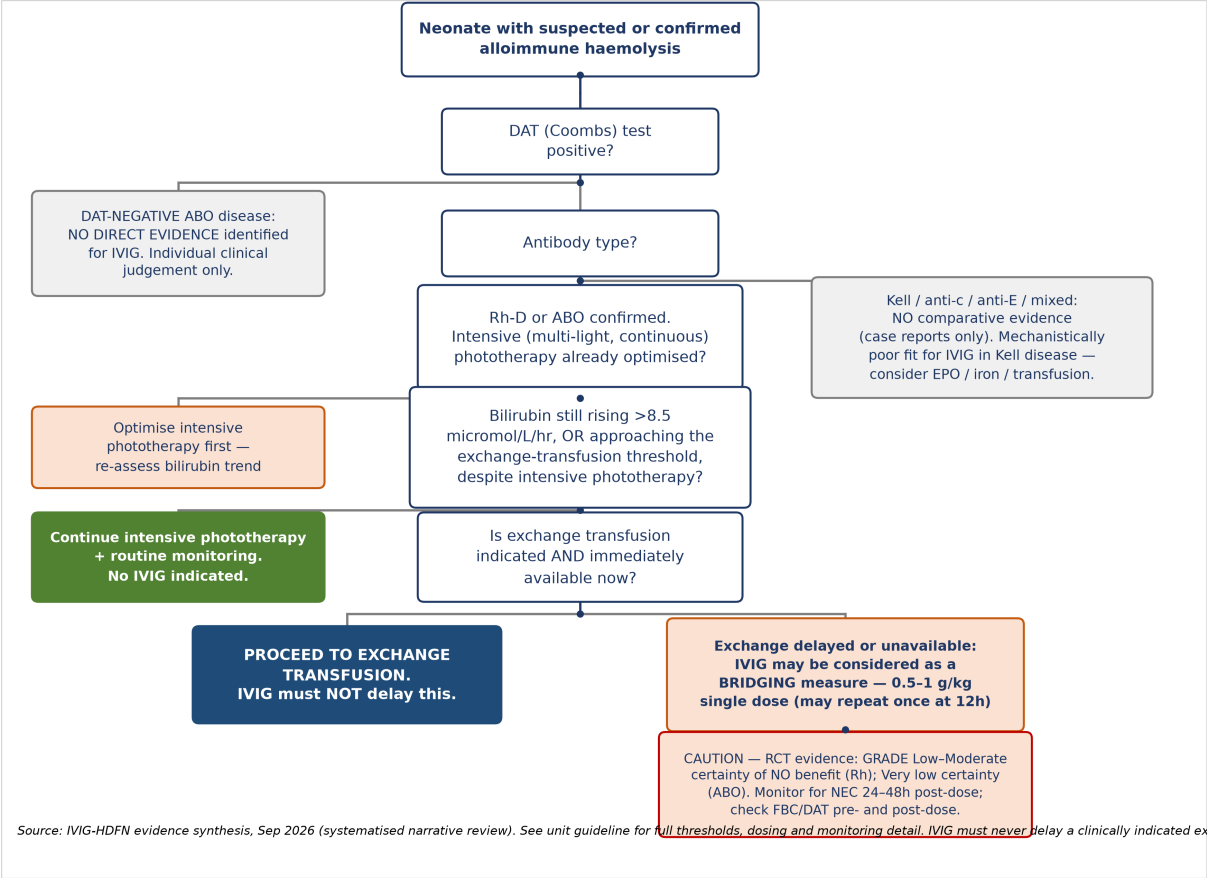
Domain B (d); Domain C (a)

2022 vs. 2010

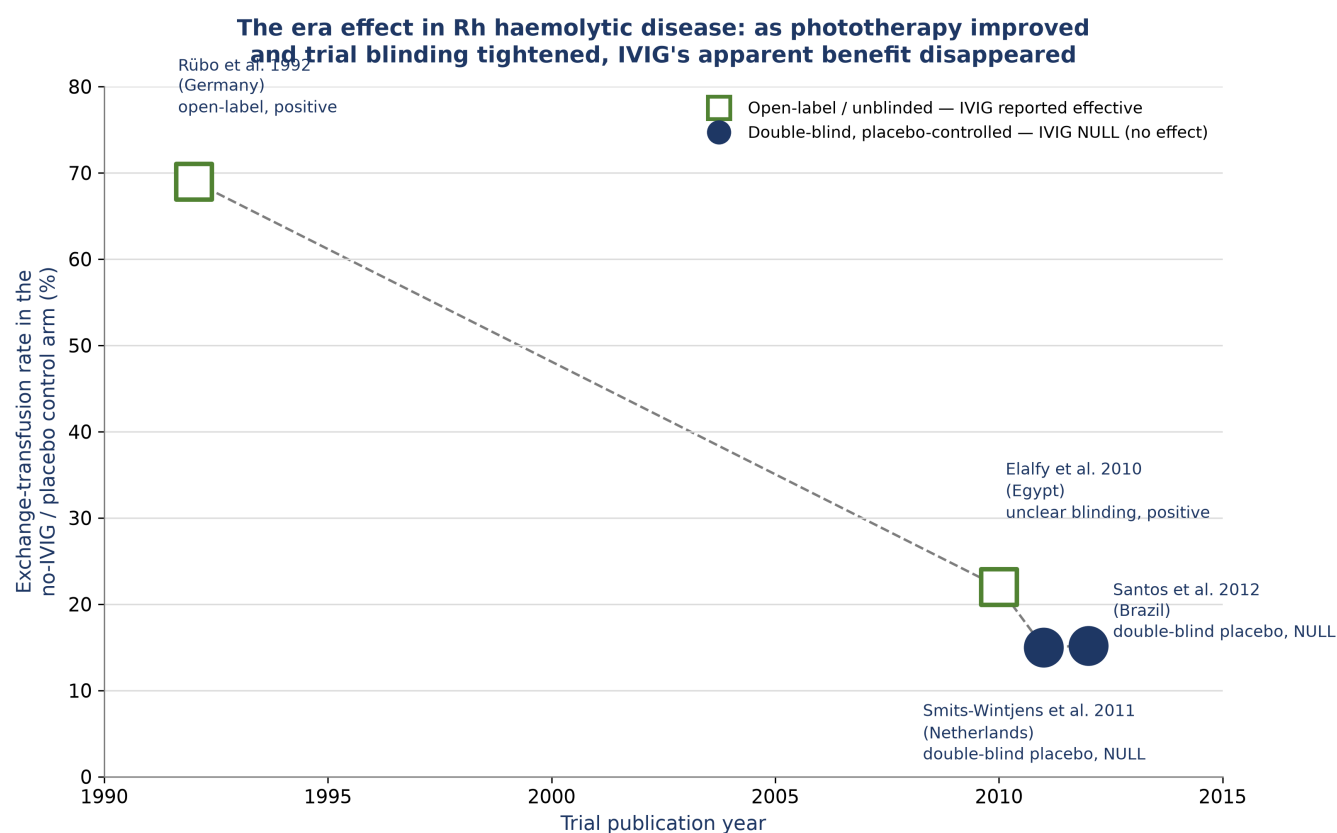
AAP and an international expert panel revised IVIG guidance downward in 2022, citing the null trials and NEC signal. NICE CG98 shows no confirmed re-evaluation since 2010.

Domain F (b), (d)

When to consider IVIG — decision pathway



Why the evidence looks contradictory: the era effect



Only the four Domain A trials reporting a numeric control-arm exchange-transfusion rate are plotted (extraction rows A1, A4, A6, A7). Every trial that also reports a large IVIG benefit is open-label or unblinded; both placebo-controlled trials are null.

Certainty of the evidence, by outcome and clinical subgroup

	Term Rh-D	Preterm Rh-D	ABO, DAT+	ABO, DAT-	Kell / anti-c / anti-E / mixed
Exchange transfusion	Low-Moderate certainty	No direct evidence	Very low certainty	No direct evidence	No direct evidence
Necrotising enterocolitis	Very low certainty	No direct evidence	Very low certainty	No direct evidence	No direct evidence
Top-up transfusion / late anaemia	Low-Moderate certainty	No direct evidence	Very low certainty	No direct evidence	No direct evidence
Isoagglutinin-mediated haemolysis	Very low certainty	No direct evidence	Very low certainty	No direct evidence	No direct evidence
Thromboembolism / hyperviscosity	No direct evidence	No direct evidence	No direct evidence	No direct evidence	No direct evidence

No direct evidence Very low certainty Low-Moderate certainty

GRADE certainty of the evidence, by outcome and clinical subgroup. Blank cells ("No direct evidence identified") mean no comparative study addresses that subgroup for that outcome at all — not that IVIG is known to be safe or ineffective there.

Key take-aways for the unit guideline group

1. **IVIG's historical evidence base does not survive contemporary trial conditions.**
The two best-conducted Rh trials, under modern intensive phototherapy, found no benefit at all.
2. **ABO disease has a weaker case than Rh disease**, not a smaller version of the same case — no low-risk-of-bias ABO trial has ever been conducted.
3. **The necrotising enterocolitis signal is real enough to monitor for, but not proven causal** — the best-controlled study found no association once confounding by indication was addressed.
4. **IVIG must never delay a clinically indicated exchange transfusion** — this is the one point of universal agreement across every guidance document and the trial evidence.
5. **DAT-negative ABO disease and non-Rh/ABO antibodies (Kell, anti-c, anti-E) are genuine evidence vacuums**, not extrapolations from the main trial base — treat them as distinct unknowns requiring individual clinical judgement.
6. **Late anaemia surveillance should be applied to all HDN infants with significant haemolysis or IUT history**, regardless of IVIG exposure — it is a feature of the disease, not the treatment.
7. **NICE CG98's evidence base should be directly re-verified** before the unit guideline defaults to its wording — international bodies revised their recommendations in 2022 in ways NICE does not appear to have matched.

Two items must be resolved before the unit guideline is ratified: (1) direct verification of NICE CG98's underlying evidence citations, and (2) direct verification of current UK product SPCs (dosing, infusion rate, isoagglutinin content) for whichever IVIG brand is actually stocked. This review's access to nice.org.uk and medicines.org.uk was blocked; the figures used throughout are WebSearch-derived and flagged accordingly in the full guideline document.